

Ruthvika Reddy Vummadi

SOFTWARE ENGINEER

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SUMMARY

- Experienced Software Engineer with 4 years in full-stack development, specializing in building secure, scalable, and enterprise-grade applications for financial services and AI-driven business solutions.
- Proven expertise in Python, Java, and JavaScript ecosystems, delivering backend microservices and modern web applications that streamline business-critical processes.
- Hands-on experience in designing RESTful and event-driven architectures, leveraging frameworks such as Django, Flask, FastAPI, Spring Boot, and React.js for high-performance solutions.
- Skilled in implementing cloud-native deployments on AWS, GCP, and Red Hat OpenShift with Docker and Kubernetes, enabling reliable, containerized application delivery.
- Background in integrating AI/ML models into production systems, including NLP-powered sentiment analysis, anomaly detection, and intelligent recommendation engines.
- Strong foundation in database design and optimization across relational and NoSQL platforms, including PostgreSQL, Snowflake, MongoDB, and DynamoDB, ensuring efficient data management.
- Experienced in DevOps practices and Agile delivery, including CI/CD pipeline automation with Jenkins, GitHub Actions, and GitLab CI/CD, coupled with continuous monitoring through ELK, Splunk, and Prometheus.
- Adept at collaborating with cross-functional teams, bridging requirements from business analysts and data scientists to deliver robust software solutions aligned with enterprise and regulatory standards.

SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, SQL

Frameworks & Libraries: Django, Flask, FastAPI, React.js, Node.js, Pandas, NumPy, PySpark, TensorFlow, Scikit-learn, Spring Boot, Hibernate, Angular 17, Next.js,

Web Technologies: HTML5, CSS3, RESTful APIs, GraphQL, WebSockets, Microservices, JSON, AJAX

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Ansible, Helm

AI & Machine Learning: AI/LLM Integration (OpenAI, LangChain), Retrieval-Augmented Generation (RAG), NLP

Databases: PostgreSQL, MySQL, MongoDB, Oracle, DynamoDB, Snowflake, Firestore, Redis

Agile & Project Tools: Agile, Scrum, JIRA, Confluence, Trello, Git, GitHub, Bitbucket, GitLab

Other Tools: Apache Kafka, RabbitMQ, PyTest, UnitTest, Selenium, Postman, JMeter, JWT, OAuth2, HTTPS, OWASP Top 10, Role-Based Access Control (RBAC), Encryption Standards, Nagios, Prometheus, Splunk, Grafana, ELK Stack

WORK EXPERIENCE

State Street Corporation

Massachusetts, USA

Software Engineer

Aug 2024 – Present

- Partnered with business analysts to define enhancements for order management and post-trade settlement modules, ensuring compliance with regulatory reporting requirements.
- Designed and optimized database schemas in Snowflake and PostgreSQL, applying partitioning and indexing strategies that reduced portfolio risk report generation from 20 minutes to under 8 minutes.
- Developed Python-based RESTful microservices (Flask, FastAPI) to connect order capture with settlement and compliance workflows, improving reconciliation efficiency by 30%.
- Led system design of microservices architecture, including API contracts, data flow diagrams, and integration patterns, enabling scalable and maintainable trade processing solutions.
- Integrated AI-driven anomaly detection models (scikit-learn, TensorFlow) into trade validation, lowering compliance false positives by 25% and enhancing data reliability.
- Implemented event-driven trade ingestion with Apache Kafka and Python consumers, enabling real-time validation of 1M+ daily trades and eliminating downstream data delays.
- Designed and developed client onboarding applications with React.js and Django, embedding KYC/AML verification that cut manual review effort by 40% and reduced onboarding time from 10 to 6 days.
- Containerized services with Docker and deployed on AWS EKS (Kubernetes), introducing autoscaling for quarter-end trading peaks and improving system availability by 35%.

- Automated CI/CD pipelines with Jenkins, Git, SonarQube, and pytest, increasing release frequency by 50% while reducing post-release incidents by 20%.

IBM

Hyderabad, India

Application Developer

Jul 2020 - Jul 2023

- Collaborated with product owners and data scientists to translate business requirements into a scalable web application applying IBM watsonx.ai NLP models for automated sentiment analysis of customer reviews
- Built the backend using Python and Django REST Framework, developing modular APIs that centralized dealer data across 200+ locations, reducing duplicate records by 20%.
- Architected the end-to-end system design, including microservices-based backend, service orchestration on Red Hat OpenShift, and API gateway configuration for secure and scalable communication.
- Integrated watsonx.ai sentiment scoring services into the workflow, classifying reviews with ~82% accuracy, which enabled faster identification of negative feedback and cut escalation times by 35%
- Designed and optimized the PostgreSQL schema with indexing and partitioning strategies, reducing API query latency by 25% and ensuring stable performance for ~10K daily requests.
- Developed a responsive React.js frontend with Redux, delivering dashboards that visualized sentiment trends, service ratings, and complaint categories, which improved adoption among dealership managers by 30%
- Defined caching, queuing, and logging strategies (Redis for caching, RabbitMQ for asynchronous processing, ELK Stack for logging/monitoring) to handle spikes in traffic and maintain reliable throughput.
- Containerized services with Docker and deployed them on Red Hat OpenShift, achieving 99.9% system availability and enabling seamless horizontal scaling during high-traffic events.
- Automated builds, testing, and deployments using Jenkins and IBM Cloud DevOps, reducing manual deployment effort by 70% and shortening release cycles from two weeks to five days.
- Engineered a secure authentication layer using OAuth 2.0 and JWT tokens, safeguarding sensitive dealer data and passing all enterprise security audits.
- Built system availability and disaster recovery mechanisms with OpenShift replicas and automated backup jobs, ensuring fault tolerance and near-zero downtime.
- Implemented API caching and load balancing to improve response times by 15% while maintaining consistent SLA compliance.
- Wrote unit tests (PyTest) and UI automation tests (Selenium), achieving ~90% coverage and lowering post-release defects by 20%, while delivering features in Agile two-week sprints.

PROJECTS

Facial Expression Authentication System

- Built a multi-factor authentication web application using React and Python, deployed on Vercel, that verifies users through facial expressions. Implemented JWT tokens and password hashing for secure login, with SQLite managing credentials and tokens. Enhanced authentication robustness while ensuring lightweight, scalable deployment.

Tech Stack: React, Python, Vercel, JWT, SQLite

AI-Powered Course Recommendation System

- Designed and implemented a full-stack course recommendation platform for Clark University, utilizing MySQL, Spring Boot (JPA, Hibernate), and RESTful APIs to manage course data across three majors. Integrated Google Gemini AI to deliver personalized recommendations with 85% accuracy, enhancing academic planning and career alignment.

Tech Stack: MySQL, Spring Boot, JPA, Hibernate, RESTful API, Google Gemini AI

EDUCATION

Master of Science in Information Technology

Clark University, Worcester, Massachusetts, United States

Bachelor in Technology in Computer Science and Engineering

TKR College of Engineering & Technology, Hyderabad, Telangana, India

CERTIFICATIONS

- Programming with JavaScript by Meta
- Microsoft Certified: Azure Fundamentals
- Generative AI – for Software developers by IBM